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MIL-S-46889B(MI)
25 July 1990
SUPERSEDING
MIL-S-46889A(MI)
2 August 1985

MILITARY SPECIFICATION

STEEL, SPRING, CHROMIUM-NICKEL ALLOY

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of chromium-nickel alloy spring steel.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

FED-STD-151 - Federal Test Method Standard Metals: Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9535
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MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-163	-	Steel Mill Products Preparation for Shipment and Storage

STANDARDS

MILITARY

MIL-STD-1190	-	Marking for Shipment and Storage Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 8	-	Tension Testing of Metallic Materials, Standard Test Methods of

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Chemical composition. The chemical composition of the steel shall be in accordance with the requirements specified in table I.

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TABLE I. Chemical composition.

Elements	Percent by weight
Carbon	0.07 to 0.11
Manganese	1.00 to 1.50
Phosphorus	0.03 maximum
Silicon	0.90 to 1.40
Sulphur	0.30 maximum
Chromium	16.00 to 16.60
Nickel	7.00 to 7.60
Molybdenum	0.60 to 0.80

3.3 Condition. Unless otherwise specified (see 6.2), the steel shall be furnished in the cold rolled condition.

3.4 Mechanical properties. The steel shall exhibit the mechanical properties specified in table II when heat treated at 426 ± 11° Celsius (C) for 4 hours ± 20 minutes.

TABLE II. Mechanical properties.

Thickness (inch)	Ultimate tensile strength (psi)	Yield strength (psi)	Elongation (Percent in 1 inch, minimum)
0.0210 or less	290,000 to 320,000	270,000 to 300,000	2 to 4

3.5 Physical properties.

3.5.1 Length, width, and thickness. The steel shall be furnished in the length, width, and thickness specified in the contract or purchase order (see 6.2).

3.5.2 Thickness tolerances. The thickness tolerances of steel furnished in widths of less than 3 inches shall be specified in table III. The thickness tolerances of steel furnished in widths of 3 inches and greater shall be specified in the contract or purchase order (see 6.2).

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TABLE III. Thickness tolerances.

Thickness (inch)	Tolerance (inch, plus or minus)
Under 0.0040	0.00016
0.0040 to 0.0063	0.00020
0.0063 to 0.0080	0.00024
0.0080 to 0.0100	0.00028
0.0100 to 0.0120	0.00032
0.0120 to 0.0160	0.00036
0.0160 to 0.0200	0.00044
0.0200 to 0.0340	0.00052

3.6 Workmanship. The steel shall be uniform in quality and condition, clean, sound, and free from internal or external imperfections detrimental to the fabrication or performance of processed parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in FED-STD-151 and 4.6 herein.

4.3.1 Lot. A lot shall consist of all steel submitted for acceptance at the same time which has been prepared without change of materials or process in one continuous period of operation.

4.3.2 Sampling. Unless otherwise specified, a sample, as specified in the contract or order (see 6.2) shall be subjected to the inspections and tests as specified herein.

4.4 First article inspection. When specified in the contract or order (see 6.2), a first article sample shall be subjected to first article inspection. Subsequent units shall not be considered for acceptance until Government approval of the first article sample has been obtained.

4.5 Quality conformance inspection. Quality conformance inspections shall be as specified in table IV.

TABLE IV. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
First article	3.1	4.4
Chemical composition	3.2	4.6.1
Mechanical properties	3.4	4.6.2
Physical properties	3.5	4.6.3
Workmanship	3.6	4.6.3

4.6 Test methods.

4.6.1 Chemical composition. Sampling and testing for chemical composition shall be in accordance with FED-STD-151, method 111.2 or 112.2 (see 3.2).

4.6.2 Mechanical properties.

4.6.2.1 Tensile test specimens. Not less than two tensile specimens shall be tested for each heat or lot of steel processed. The test specimen shall conform to ASTM E 8.

4.6.2.2 Ultimate tensile strength, yield strength, and elongation. After heat treatment (see 3.4), the ultimate tensile strength, yield strength,

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and elongation shall be determined in accordance with ASTM E 8, to determine conformance to table II.

4.6.3 Physical properties. Dimensional and visual inspection of samples shall be conducted to determine compliance with the requirements specified in 3.5.1, 3.5.2, and 3.6.

4.7 Inspection of packaging. The preservation, packaging, packing, and marking shall be examined to determine compliance with Section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be level A, B, or C (see 6.2).

5.1.1 Level A or B. Preservation and packing shall be in accordance with MIL-STD-163. Marking shall be in accordance with MIL-STD-129.

5.1.2 Level C. Level C shall be in accordance with MIL-STD-1190.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The steel covered by this specification is intended for use in the fabrication of clock and gyro springs.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Whether first article is required (see 3.1 and 4.4).

d. Condition, if other than specified (see 3.3).

e. Length, width, and thickness of steel required (see 3.5.1).

f. Thickness tolerances of steel in widths of three inches and greater (see 3.5.2).

g. Levels of preservation, packing, and marking (see Section 5).

h. Sample as specified (see 4.3.2).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number

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of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this document.

6.5 Subject term (keyword) listing.

Alloys, iron
Castings, metal
Forgings
Metal strips, compressed

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing Activity
Army MI

Project No. 9535-A037

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-46889B(MI)	900725

3. DOCUMENT TITLE

STEEL, SPRING, CHROMIUM-NICKEL ALLOY

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U.S. Army Missile Command	(205) 876-6980	746-6980
c. ADDRESS (Include Zip Code) Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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